

SPY SENTINEL AI

An Alpaca-connected paper trading agent built to govern not just trades — but what evidence is allowed to teach the system.

OPTIONS ALPHA AGENTS

ALPACA

FAIL-CLOSED

TWO SEPARATE AUTHORITIES

1 EVIDENCE AUTHORITY

Is the evidence trustworthy enough to become learning?

2 PROMOTION AUTHORITY

Is independently validated learning strong enough to replace the Champion?

Execution capability demonstrated in Alpaca paper mode. Profitable edge is not claimed as proven.

What a judge can verify now

Verified frozen checkpoint from the working system- not a simulated learning loop.

FRESH PAIRED OBSERVATIONS

83

SPY + current option evidence

QUALIFIED OUTCOMES

57

10 required → READY

UNATTENDED SPY OBSERVATIONS

46

Background evidence collection

CHAMPION CHANGES

0

Self-promotion disabled

COLLECTOR PAPER ORDERS

0

Execution disabled by design

COLLECTOR LIVE ORDERS

0

Execution disabled by design

HISTORICAL RECORDS QUARANTINED

46

V196 option context unverified

PROMOTION GATE

LOCKED

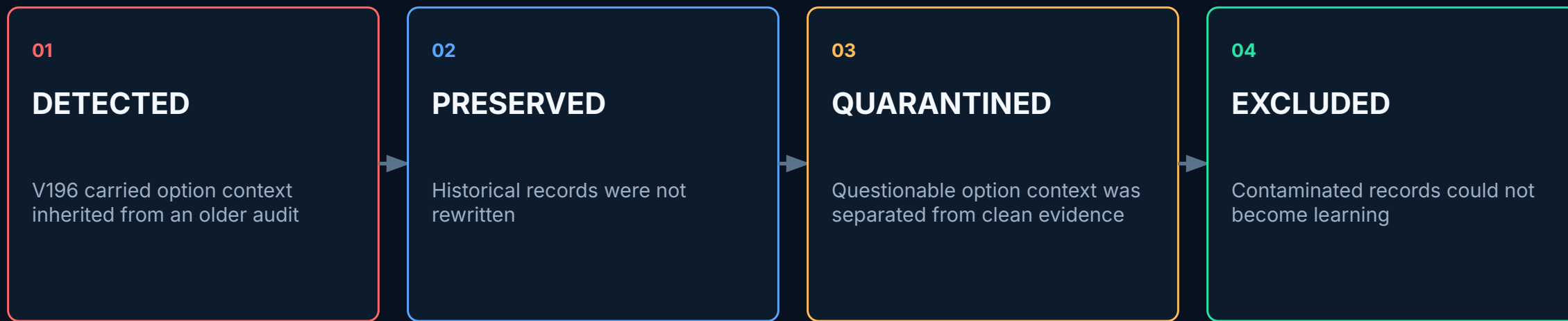
Independent validation required

CORE CLAIM

SPY Sentinel does not simulate a governance story for the demo. It shows evidence it collected, evidence it rejected, what became learning, and why that learning still cannot promote itself.

A mistake became proof of the control system

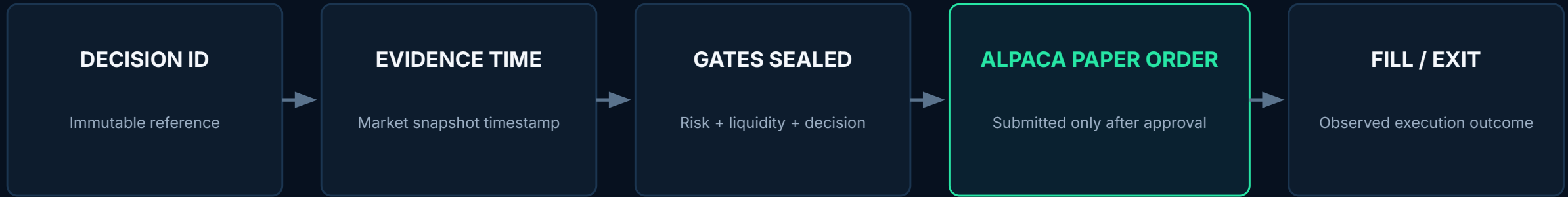
Fresh SPY prices were legitimate. Option freshness in an earlier collector could not be independently proven.



Prospective correction: V199/V201 switched to independently timestamped Alpaca option quotes. The system preserved the evidence against itself instead of making history look cleaner than it was.

Prove the chronology before execution

Reasoning is useful only if a judge can see it existed before the order — not reconstructed afterward.



EXECUTION REALITY

Controlled Alpaca paper execution, monitoring and deterministic exits have been demonstrated separately. Real fills can differ from quoted midpoint because of spread, slippage, latency and liquidity.

Learning has to pass two different boundaries

A Challenger being “ready” is not the same thing as being authorized to replace production behavior.

EVIDENCE AUTHORITY

Observe

Real market evidence



Evaluate

Evidence quality + risk



Experience Log

Decision + provenance



Outcome Label

Observed or counterfactual



Quarantine

Exclude defective evidence

PROMOTION AUTHORITY

Challenger

Training-ready evidence



Independent Validation

Separate promotion gate



Champion

Protected unless gate passes

SELF-PROMOTION: DISABLED

Credibility before victory laps

The proof package separates demonstrated capability from evidence of a repeatable trading edge.

DEMONSTRATED

- Alpaca-connected paper lifecycle
- Deterministic risk gates
- Monitoring + exit logic
- Audit trail + mistake quarantine
- Governed Challenger readiness

OBSERVED

- Paper fills and exits
- Counterfactual option outcomes
- Training-ready evidence counts

Observed paper outcome — not, by itself, proof of repeatable edge.

NOT CLAIMED

- Proven profitable edge
- Guaranteed execution at midpoint
- Authorization for live money
- Automatic Champion replacement

The system can say NO — to trades, to evidence, and to its own learning.

Verify the system in minutes

The public package is intentionally ordered around proof, not architecture theater.

1**WATCH****2:42 final demo**

See the working proof story end-to-end

2**OPEN****Live Judge Proof**

Inspect evidence counts, quarantine and promotion status

3**VERIFY****GitHub proof package**

README banner, snapshots, verifier and walkthrough

4**TRACE****Decision chronology**

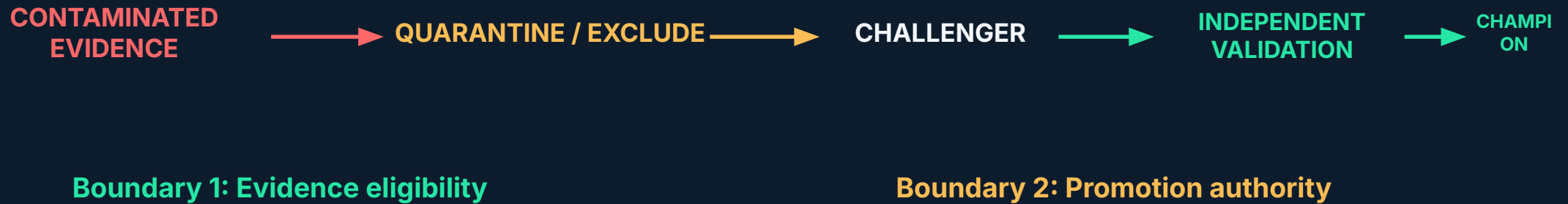
Decision ID → evidence time → gates → paper order → fill/exit

Public proof: ajennings1974.github.io/SPY-Sentinel-AI/

GitHub: github.com/ajennings1974/SPY-Sentinel-AI

The deeper trust boundary

Many agents can reject a trade. SPY Sentinel asks a harder question: what is allowed to become learning, and who is allowed to promote it?



No Champion change occurred. That is not an unfinished demo — it is the promotion firewall doing its job.

SPY Sentinel AI • Real evidence • Controlled paper execution • Fail-closed learning